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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 23	
TITLE : JDBC ResultSet	

Problem Statement

Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.

Learning Objectives

To navigate and retrieve records using ResultSet methods.

Programme



```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
import java.sql.SQLException;

public class JDBCResultSet {

    Run | Debug
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "Husam123";

        try {
            Connection con = DriverManager.getConnection(
                url, user, password
            );

            System.out.println("Database connected successfully.");

            Statement stmt = con.createStatement();

            String query = "SELECT * FROM student";
            ResultSet rs = stmt.executeQuery(query);

            System.out.println("\nStudent Records:");
            System.out.println("--------------------------------");

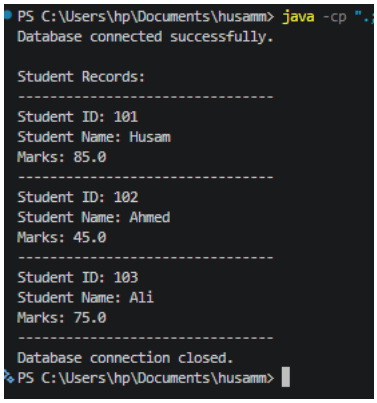
            while (rs.next()) {
                System.out.println("Student ID: " + rs.getInt("id"));
                System.out.println("Student Name: " + rs.getString("name"));
                System.out.println("Marks: " + rs.getDouble("marks"));
                System.out.println("--------------------------------");
            }

            rs.close();
            stmt.close();
            con.close();

            System.out.println("Database connection closed.");

        } catch (SQLException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Output



```
PS C:\Users\hp\Documents\husamm> java -cp ".
Database connected successfully.

Student Records:
--------------------------------
Student ID: 101
Student Name: Husam
Marks: 85.0
--------------------------------
Student ID: 102
Student Name: Ahmed
Marks: 45.0
--------------------------------
Student ID: 103
Student Name: Ali
Marks: 75.0
--------------------------------
Database connection closed.
PS C:\Users\hp\Documents\husamm>
```

Exercise 1. Create a program to display database records one by one using ResultSet.

Code

TelemedicineBilling.java | SmartLogistics.java | Untitled-2 | JDBCSelect.java | Exercise1.java

Exercise1.java > ...

```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.sql.ResultSet;
import java.sql.SQLException;

public class Exercise1 {
    Run | Debug
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "Husam123";

        try {
            Connection con = DriverManager.getConnection(
                url, user, password
            );

            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");

            System.out.println("Student Records:");

            while (rs.next()) {
                System.out.println("ID: " + rs.getInt("id"));
                System.out.println("Name: " + rs.getString("name"));
                System.out.println("Marks: " + rs.getDouble("marks"));
                System.out.println("--------------------");
            }

            rs.close();
            stmt.close();
            con.close();

        } catch (SQLException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

output

```
PS C:\Users\hp\Documents\husamm> java -cp
Student Records:
ID: 101
Name: Husam
Marks: 85.0
----------------------
ID: 102
Name: Ahmed
Marks: 45.0
----------------------
ID: 103
Name: Ali
Marks: 75.0
----------------------
PS C:\Users\hp\Documents\husamm>
```

2. Create a Java program that retrieves employee records from a database and displays them one by one using ResultSet. Show employee ID, name, department, and salary sequentially.

Code

```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.sql.ResultSet;
import java.sql.SQLException;

public class Exercise2 {

    Run | Debug
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "Husam123";

        try {
            Connection con = DriverManager.getConnection(
                url, user, password
            );

            Statement stmt = con.createStatement();

            String query = "SELECT * FROM employee";
            ResultSet rs = stmt.executeQuery(query);

            System.out.println("Employee Records:");
            System.out.println("-----------------------------");

            while (rs.next()) {
                System.out.println("Employee ID: " +
                    rs.getInt("employee_id"));

                System.out.println("Name: " +
                    rs.getString("name"));

                System.out.println("Department: " +
                    rs.getString("department"));

                System.out.println("Salary: " +
                    rs.getDouble("salary"));

                System.out.println("-----------------------------");
```

Output

```
PS C:\Users\hp\Documents\husamm> java
Enter Login ID: DOC101
Enter Password: DOC101
Authentication Failed!
Invalid Login ID or Password.
Access Denied.
PS C:\Users\hp\Documents\husamm>
```

Conclusion

The program successfully demonstrates the use of JDBC ResultSet to navigate through and retrieve records from a MySQL database. It uses methods such as next(), getInt(), and getString() to access and display database records sequentially.